PE3101P: Decision and Social Choice

Lecture 2 — Probability

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20 August 2026

1 Logistics

1.1 The week 1 exercise sheet is up

Self-explanatory. See my website, `https://tomifrancis.com/pe3101p.html`

1.2 New document: set theory primer

I'm giving you the greatest gift a person can receive: extra homework.

It's optional, but very helpful if you find yourself getting confused by all these sets and intersections and unions and so on.

Available on my website in pdf and html formats. The html version is recommended, as it includes features that can't easily appear in pdf format.

1.3 First for-credit assignment

This will go out on my website over the weekend. It is due on **Friday of Week 4 (September 4th)**, by midnight.

1.4 My office

I'm moving office today. From now on I'm in **AS3-05-01**.

Office hours today are 16:30–18:00, later than usual, because I'm moving in right after this lecture.

1.5 Register

We've got a printed version this time. Find your name, sign. It's in alphabetical order. If you can't find yourself, I don't have records of you yet. Add your name and student number, then sign.

2 Recap of last week

2.1 Fact-relative and evidence-relative correctness

- What you ought to do in the **fact-relative sense** is what will *in fact* be best/best serve your preferences.
- What you ought to do in the **evidence-relative sense** is what *your evidence indicates* will be best, or will best serve your preferences, on the balance of probabilities.
- If you run across the road without looking and don't get hit by a car, what you did was correct in the fact-relative sense, but likely a mistake in the evidence-relative sense.

- In decision theory, we're interested in good decision-making in the evidence-relative sense.

2.2 Acts, states of nature, outcomes

	Miners in A	**Miners in** B
Block A	all 10 live	all 10 die
Block B	all 10 die	all 10 live
Block neither	1 dies	1 dies

- A decision table, as above, is a model for describing a decision situation. It gives information about the *acts*, *states of nature* and *outcomes* involved in a decision situation.
- The acts are the different decisions the agent could make.
- The states of nature are the ways the world could be, which the agent is uncertain about.
- The outcomes are the ways things could turn out, depending on how the world is and what the agent chooses.

2.3 Conditions on states of nature

- We introduced two conditions on states of nature:

(i) Together with the act chosen, they resolve the uncertainty inherent in the decision situation and guarantee that one particular outcome results.

(ii) The act chosen by the agent does not affect whether the state of nature will occur, or what its probability is.

In addition, we will see two more conditions today which apply to the set of states of nature (rather than individual states):

(iii) They are *mutually exclusive* - at most one can actually happen.

(iv) They are *jointly exhaustive* - at least one must happen.

- Therefore: *exactly* one will happen, just like if you roll a die, it shows one particular number face up (provided nothing goes wrong with the roll).

We will have the technology to state this more precisely later on.

2.4 Chance and credence

- Which act it is sensible to select depends on how likely you think each state of nature is to obtain. We call this the *credence* you assign to that state of nature.
- Credences are a lot like chances, and we call them probabilities, but there are some kinds of chances, called objective chances, which are different to credences.
- Objective chances are out there in the world, credences are in the agent's head.
- Objective chances could be the frequency with which a type of event actually happens, or a probability output of a scientific theory like quantum mechanics.
- Credences are, roughly, how likely an agent thinks an event is, on the balance of the evidence available to her.

2.5 Utility and expected utility

A fair coin. Take the bet and you get \$100 on heads and nothing on tails; decline and you get \$50 either way.

	Heads ($p = 0.5$)	**Tails** ($p = 0.5$)
Take the bet	100	0
Decline	50	50

- Utilities are numbers representing how good an outcome is by the agent's lights.
- Where do these numbers come from? We'll get to that in later weeks. For now, just remember that more is better.
- If utilities have been assigned to outcomes, the expected utility of an act is the sum of the utilities of the outcomes, weighted by their probabilities; i.e., the probabilities of the states of nature in which those outcomes occur.

2.6 Dominance

	Miners in A	**Miners in** B
Block A	all 10 live	all 10 die
Block B	all 10 die	all 10 live
Block neither	1 dies	1 dies
Block both	all 10 live	all 10 live

One act weakly statewise dominates another if it is better in some state of nature and at least as good in the rest.

	Miners in A	**Miners in** B
Block A	10	0
Block B	0	10
Block neither	9	9
Block both	10	10

One act strongly statewise dominates another if it is better in all states of nature.

2.7 Bayesianism: the two constraints

- The main topic for today. We're going to discuss two alleged requirements rationality imposes on your credences.
- Probabilism: the conditions (AKA the Kolmogorov axioms) we saw last week. (We'll come back to them, don't worry!)
- The principle of conditionalisation: a principle governing how your credences should change in response to new evidence.

3 Probabilities: the basics

We begin with a *sample space* Ω, which is just a collection of all of the possible outcomes or states of nature over which we are uncertain in the given situation.

For example, it could be the possible results of rolling a die, or coin flips, different ways the weather could go, or the decisions somebody else is going to make without your input.

A sample space is just the set of all of these outcomes. By a "set", I just mean a collection of other things. I'll have to talk about these quite a lot, so you may want to go and take a look at the *set theory primer* now available on my website if you want to get more familiar with them.

A set of some of the outcomes is called an event. For example, an event could be the die rolling 2 *or* 3, or it could be the weather being rainy *or* overcast. An event can also contain just a single outcome, and in this case although an event is *formally* distinct from an outcome, we can treat them like they're the same time. (For example, the event that the die rolls 1).

An event can also be that *any* of the possible outcomes happens. In this case, it is equal to the sample space Ω. This is why I called it the *universal event* last week. The sample space and the universal event are the same thing: the set of all of the things that could happen.

The set containing no outcomes at all is also an event, albeit one that can never happen. We denote this by $\varnothing$ and we can call it the empty event.

3.1 Combining events

If P and Q are events, we can write $P \cup Q$ for the event that something in either P *or* Q (or both) happens. This is a set-theoretic union. But, if you like, you can also write $P \vee Q$. Technically this is an abuse of notion (more on that later), but we won't worry about it.

Similarly, for events P and Q we can write $P \cap Q$ for the event that something in *both* P and Q happens. This is a set-theoretic intersection. Similarly to before, you can write $P \wedge Q$ or $P \& Q$ instead if you like.

Finally, any event P has a complement P^c, which we will often denote by $\neg P$, which is the event that contains all the outcomes which are in the sample space but not in P.

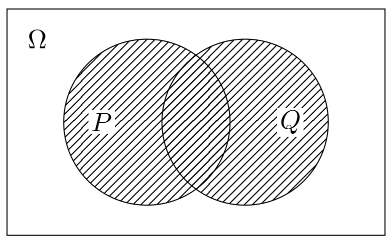

Figure 1: The union $P \cup Q$: everything in P, in Q, or in both.

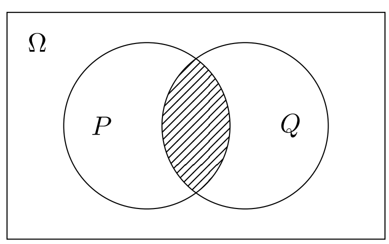

Figure 2: The intersection $P \cap Q$: everything in both P and Q.

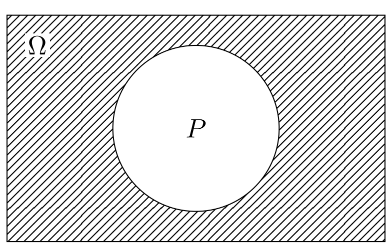

Figure 3: The complement P^c: everything in Ω that is not in P.

3.2 A note on this notation

In general, and officially, I take the sample space to be a set of outcomes, and events to be *subsets* (sets which might be smaller) of those outcomes. The proper way to combine these kinds of events to get other events is to use the set theoretic notions of union and intersection. The union of sets X and Y is the set that contains everything in *either* X or Y, and the intersection of X and Y is the set that contains everything in *both* X and Y. (For much more detail on this, see the primer.)

Other authors, including Schwarz in BDRC, prefer to talk about propositions, and to use the logical notions of conjunction ($\wedge$) and disjunction ($\vee$)—I just mean "and" and "or".[1]

Technically, these are different. "And" and "or" are truth-functional conditions which take truth-apt objects like sentences or propositions and return further sentences or propositions whose truth values are grounded in the truth values of their component parts. "Intersection" and "union" are set-theoretic operators which combine two sets to form another set.

They do different things. But they are very tightly related and behave exactly the same way in most contexts, so in this course we will mostly ignore the differences between them. In other words, I'm not going to penalise you if you write $\wedge$ instead of $\cap$.

3.3 A note on non-measurability

Generally in mathematics a *Probability Space* comes with a sample space Ω and a specification of events Σ—a list of all those subsets of Ω which "count" as events. (It also has what's called a *probability measure*, which we're just about to talk about.)

The reason why the specification of events Σ has to exist in general is that *in general, not every subset of a sample space can coherently be assigned a probability*. The reasons why are quite complicated. For our purposes, we can ignore the problem, and we can generally assume that *any* set of outcomes counts as an event.

3.4 Mutually exclusivity, joint exhaustiveness

Sometimes, two events have no outcome in common at all. When events X and Y are like this, we say that they are *mutually exclusive* or *disjoint*, and the condition for it is that $X \cap Y = \varnothing$, the empty event.

Sometimes, two or more events are such that it is guaranteed that at least one will happen. In this case, we say that the events are *jointly exhaustive*, and the condition for it is that $X_1 \cup \cdots \cup X_n = \Omega$, i.e., every outcome is in at least one of the events in question.

And when a set of events are mutually exclusive and jointly exhaustive, we say that they *partition* the sample space.

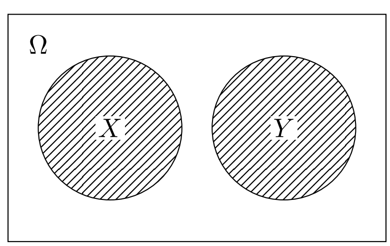

Figure 4: Mutually exclusive (disjoint) events: $X \cap Y = \varnothing$.

[1] Strictly speaking, since Schwarz characterises propositions as sets of possible worlds, these actually amount to exactly the same thing as the set-theoretic notions of intersection and union anyway.

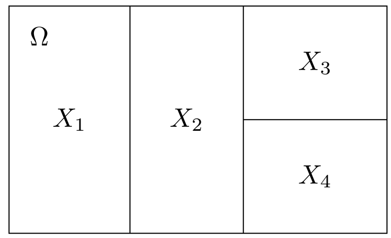

Figure 5: A partition of Ω: the cells are pairwise disjoint *and* jointly exhaustive, so exactly one of them obtains.

Remember from earlier: we want to also require that the set of states of nature is mutually exclusive and jointly exhaustive. So, the states of nature partition the sample space for each act.

4 Probabilism

4.1 Assigning Probabilities

We want to talk about assigning probabilities to events, and what are the constraints on doing so. In particular, we're interested in the structure of credence: the constraints placed by epistemic rationality on how our credences should look.

4.2 Epistemic irrationality

Intuitively, some patterns of credence we can have are incorrect.

Have a think about these, and then tell me what's wrong with having credences like this.

Example 1. Joe thinks that there is a 50% chance that it will rain today and a 20% chance that it will not rain today.

Example 2. Biqing thinks that there is a 60% chance she failed the exam she just took, and a 70% chance she failed it badly.

Example 3. Homer thinks that there's a 30% chance he'll drink one beer, a a 50% chance he'll drink more than one beer, and a 70% chance he'll drink at least one beer.

4.3 The Probability Axioms (again)

Lecture 1 stated these informally:

(i) **Non-negativity.** Probabilities can't be negative.

(ii) **Normalisation.** The probability of the universal event—of *something* happening—is 1.

(iii) **Additivity.** The probability of *either* of two completely separate events happening is the sum of their probabilities. (E.g., the probability of a rolled die landing 1 *or* 2 is $1/6 + 1/6 = 1/3$.)

We now have names for all the pieces. Ω is the sample space, Σ is the set of events, and $\mathrm{Cr}(\sigma)$ is the credence assigned to an event $\sigma \in \Sigma$. "Completely separate" is disjointness, which we defined above as $X \cap Y = \varnothing$.

(i) **Non-negativity:** $\mathrm{Cr}(\sigma) \geq 0$ for every $\sigma \in \Sigma$.

(ii) **Normalisation:** $\mathrm{Cr}(\Omega) = 1$.

(iii) **Finite additivity:** if $X \cap Y = \varnothing$, then $\mathrm{Cr}(X \cup Y) = \mathrm{Cr}(X) + \mathrm{Cr}(Y)$.

More generally, if the events $\sigma_0, \sigma_1, \ldots$ are pairwise disjoint, then $\mathrm{Cr}(\bigcup_i \sigma_i) = \mathrm{Cr}(\sigma_0) + \mathrm{Cr}(\sigma_1) + \ldots$

5 Conditional probabilities and Conditionalisation

5.1 The definition

Definition (Conditional probability)**.**

$$\mathrm{Cr}(A \mid B) = \frac{\mathrm{Cr}(A \cap B)}{\mathrm{Cr}(B)} \qquad \text{provided } \mathrm{Cr}(B) > 0.$$

5.2 Why that formula

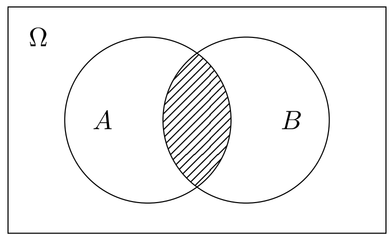

Figure 6: Before conditioning: Ω is everything, and the shaded region $A \cap B$ is measured against all of it.

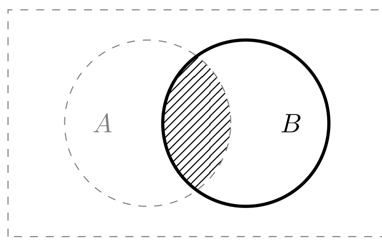

Figure 7: After conditioning on B: everything outside B is discarded, and the same shaded region is now measured against B alone.

5.3 Conditionalisation

Principle 1 (Conditionalisation)**.** *Suppose your credences are given by* Cr*, and then you learn* E *and nothing more. Your new credence in any event* A *should be your old credence in* A *conditional on* E*:*

$$\mathrm{Cr}_{new}(A) = \mathrm{Cr}(A \mid E).$$

5.4 A visual explanation of conditionalisation

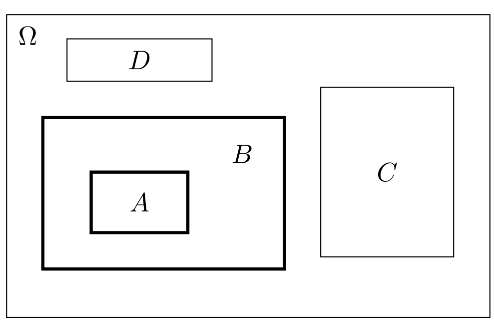

Figure 8: Stage 1. The sample space Ω, with areas representing probabilities. B is the event we are going to condition on, and A lies wholly inside it.

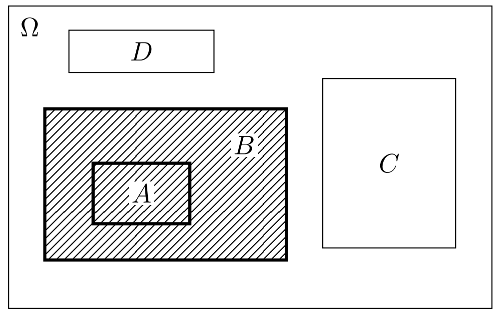

Figure 9: Stage 2. The conditioning event B shaded.

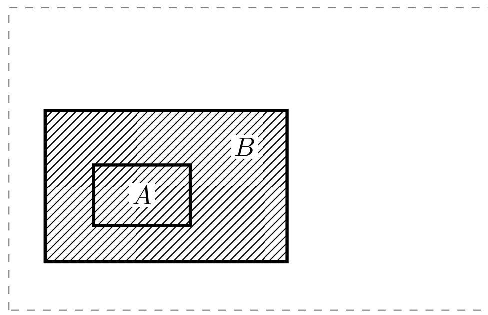

Figure 10: Stage 3. Everything outside B is discarded. Nothing has moved and nothing has changed size yet.

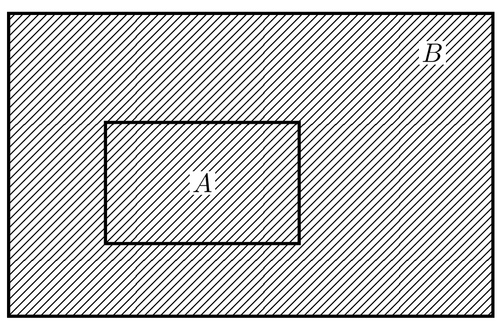

Figure 11: Stage 4. B blown up until it fills the space Ω used to. The scaling is uniform, so every ratio of areas inside B is unchanged.

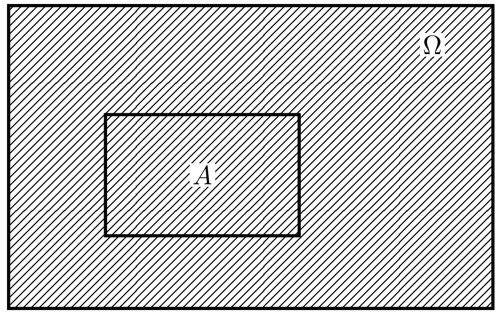

Figure 12: Stage 5. Relabel: B is the new sample space. The credence in A after conditioning on B is the fraction of this picture that A occupies.

5.5 The multiplication rule and total probability

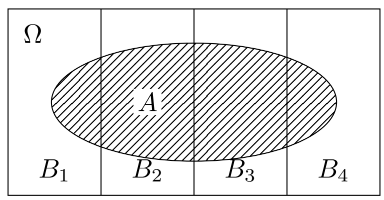

Figure 13: The law of total probability. The B_i partition Ω, so A is cut into the disjoint pieces $A \cap B_i$, and their probabilities sum to $\mathrm{Cr}(A)$.

5.6 Bayes' theorem

What we have, straight from the definition, is a pair of equations:

$$\mathrm{Cr}(A \mid B) = \frac{\mathrm{Cr}(A \cap B)}{\mathrm{Cr}(B)} \qquad \text{and} \qquad \mathrm{Cr}(B \mid A) = \frac{\mathrm{Cr}(A \cap B)}{\mathrm{Cr}(A)}.$$

What we want is a way of getting from one conditional probability to the other:

Principle 2 (Bayes' theorem)**.**

$$\mathrm{Cr}(A \mid B) = \frac{\mathrm{Cr}(B \mid A)\ \mathrm{Cr}(A)}{\mathrm{Cr}(B)} \qquad \textit{provided } \mathrm{Cr}(A), \mathrm{Cr}(B) > 0.$$

Sometimes you know one of these conditional probabilities and want the other, and Bayes' theorem is what gets you across. That happens often enough to be worth having, though not always.

Exercise 1. Prove Bayes' theorem

5.7 Bayes factors

Suppose you are weighing two hypotheses, H_1 and H_2, and some evidence E arrives. The *Bayes factor* is

$$\frac{\mathrm{Cr}(E \mid H_1)}{\mathrm{Cr}(E \mid H_2)}.$$

It measures how much better E is explained by H_1 than by H_2, and it is the whole of what the evidence contributes. That is easiest to see in odds form:

$$\underbrace{\frac{\mathrm{Cr}(H_1 \mid E)}{\mathrm{Cr}(H_2 \mid E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathrm{Cr}(E \mid H_1)}{\mathrm{Cr}(E \mid H_2)}}_{\text{Bayes factor}} \times \underbrace{\frac{\mathrm{Cr}(H_1)}{\mathrm{Cr}(H_2)}}_{\text{prior odds}}$$

So the Bayes factor is exactly the multiplier: it says how far the evidence moves you, and your prior says where you started from. A Bayes factor of 1 means the evidence is equally expected either way, and moves you not at all.

5.8 Examples, part 1

Exercise 2. A fair six-sided die is rolled. What is your credence that it landed 6, given that it landed even?

Exercise 3. Two fair coins are tossed. What is your credence that both landed heads, given that at least one did? (The answer is not $1/2$.)

Exercise 4. An urn holds 3 red balls and 7 black. Two are drawn without replacement. What is your credence that the second is red, given that the first was red? And what is your credence that the second is red, before you look at the first?

Exercise 5. $\mathrm{Cr}(A) = 0.5$, $\mathrm{Cr}(B) = 0.4$, and $\mathrm{Cr}(A \cap B) = 0.2$. Find $\mathrm{Cr}(A \mid B)$ and $\mathrm{Cr}(B \mid A)$. Are A and B independent?

5.9 Examples, part 2

Example 4 (Medical screening)**.** A disease affects 1 person in 1000. A test never misses it: everyone who has it tests positive. But 5% of healthy people also test positive. You test positive. How worried should you be?

Out of 100,000 people: 100 have the disease and all 100 test positive; of the remaining 99,900, five per cent — that is 4,995 — also test positive. So 5,095 people test positive and only 100 of them are ill:

$$\mathrm{Cr}(\text{ill} \mid \text{positive}) = \frac{100}{5095} \approx 0.02.$$

The test is strong evidence — it multiplied the odds by about 20 — and you are still very probably fine.

Example 5 (The prosecutor's fallacy)**.** A DNA sample from the scene matches the defendant. The match probability for an innocent person is 1 in a million, and the defendant matches. The prosecution says: so there is only a one in a million chance he is innocent.

But suppose the pool of people who could have done it numbers ten million. Then about 10 innocent people match, as well as the guilty one, so roughly 1 in 11 of the people who match is guilty:

$$\mathrm{Cr}(\text{guilty} \mid \text{match}) \approx \frac{1}{11} \approx 0.09.$$

$\mathrm{Cr}(\text{match} \mid \text{innocent})$ and $\mathrm{Cr}(\text{innocent} \mid \text{match})$ are not the same quantity, and Bayes' theorem is what relates them.

Exercise 6. In the screening example, suppose the disease affects 1 person in 100 instead. Recompute $\mathrm{Cr}(\text{ill} \mid \text{positive})$. What has changed, and what has not?

6 Probabilistic independence

6.1 The definition

Definition (Probabilistic independence)**.** Events A and B are *probabilistically independent* just in case

$$\mathrm{Cr}(A \cap B) = \mathrm{Cr}(A) \cdot \mathrm{Cr}(B).$$

Where $\mathrm{Cr}(B) > 0$, this is the same as saying

$$\mathrm{Cr}(A \mid B) = \mathrm{Cr}(A),$$

which is the more intuitive form: learning B tells you nothing about A.

6.2 Independence is not mutual exclusivity

These two are easy to run together, and they are close to opposites.

Suppose A and B are mutually exclusive, and both have positive credence. Then $A \cap B = \varnothing$, so $\mathrm{Cr}(A \cap B) = 0$, while $\mathrm{Cr}(A) \cdot \mathrm{Cr}(B) > 0$. So they are *not* independent.

In fact they are as dependent as two events can be: learning that A happened takes your credence in B straight to zero.

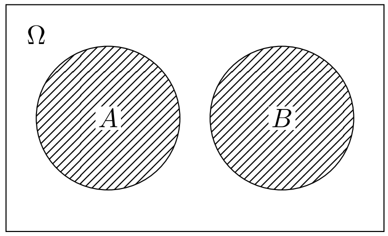

Figure 14: Mutually exclusive: A and B cannot both happen, so learning that A happened rules B out entirely. This is the opposite of independence, not an instance of it.

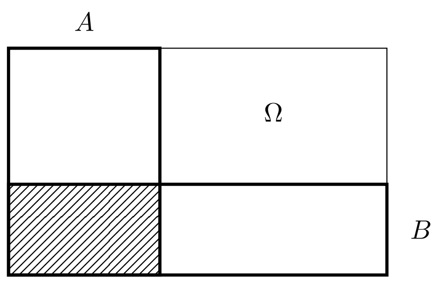

Figure 15: Independent: B takes up the same fraction of A as it does of Ω. The shaded cell is $A \cap B$.

7 Dutch book arguments

7.1 The betting interpretation of credences

We need some link between what an agent believes and what she does. The standard one, for these arguments, is betting behaviour.

Write a *\$S bet on A* for a deal that pays \$S if A is true and nothing otherwise.

Principle 3 (The betting interpretation)**.** *An agent whose credence in A is p will buy a \$S bet on A for any price below \$pS, and will sell one for any price above \$pS.*

So her credence fixes a price, and she is willing to take either side of it. We will assume this for the time being. It is a substantial assumption, and it is probably false.

7.2 The Dutch book theorem

Two of the axioms can be argued for with a *single* bet. These are the easy cases, and they are worth doing first because nothing in them depends on how a sequence of offers is handled.

Non-negativity. Suppose $\mathrm{Cr}(A) = -0.5$. By the betting interpretation she will sell a \$100 bet on A for any price above −\$50 — and a negative price means she pays. So she will happily pay someone \$40 to take the bet on, and then owes them \$100 if A turns out true.

	A	$\neg A$
Her net gain	$-\$140$	$-\$40$

She loses \$40 whatever happens, from one transaction.

Normalisation. Suppose $\text{Cr}(\Omega) = 0.9$. She will sell a \$100 bet on Ω for \$90. But Ω is guaranteed, so she collects \$90 and pays out \$100.

	Ω (certain)
Her net gain	$-\$10$

And if instead $\text{Cr}(\Omega) = 1.2$, she buys the same bet for \$110 and collects \$100, losing \$10 again.

These cases are easy, because they can be done in a single bet. Finite additivity is not so easy, because it depends on how credences relate to other credences, and so multiple bets need to be taken.

7.3 Finite Additivity Dutch Book for Myopic Choice

Suppose, as an example of a finite additivity violation, that our agent has credence 0.2 in P, credence 0.2 in Q, but credence 0.8 in $P \cup Q$.

She is offered three bets in turn, each for a moment and then withdrawn.

Bet 1 She sells a \$100 bet on P for $\$20 + \varepsilon$. Her price for that bet is \$20, so selling above it is a gain.

Bet 2 She sells a \$100 bet on Q for $\$20 + \varepsilon$. Likewise.

Bet 3 She buys a \$100 bet on $P \cup Q$ for $\$80 - \varepsilon$. Her price for that bet is \$80, so buying below it is a gain.

Each bet, taken on its own, is one she wants. So it looks as though she should take all three. But if she does, she is down about \$40 however things turn out.

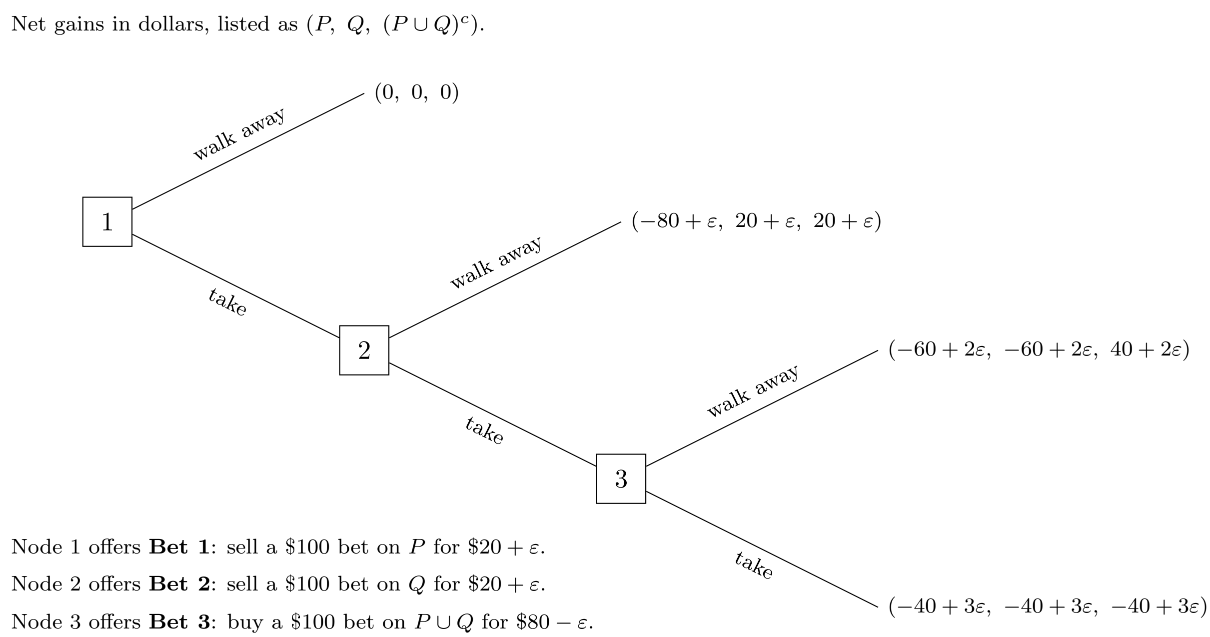

Figure 16: The Dutch book for finite additivity.

7.4 Sophisticated choice

So far we have quietly assumed that she treats each offer as if it were the only decision she will ever face. That is called *myopic* choice. An agent who instead looks ahead, and takes into account what she will do later, is called *sophisticated*. (Much more on this in week 5.)

A sophisticated agent will obviously not take all three bets. For suppose she did. Then she would go on taking bets at node 2 as well, and so at node 1 her choice would really be between

$$(0,\ 0,\ 0) \qquad \text{and} \qquad (-40+3\varepsilon,\ -40+3\varepsilon,\ -40+3\varepsilon),$$

and she would obviously take the first. So she declines at node 1.

It turns out that a sophisticated agent here makes one trade and then walks away, though we will not go through that now.

7.5 Can the book be repaired?

It can. The Dutch book for finite additivity can be reconstructed so that it works even against an agent who uses sophisticated choice — which is what she ought to be using. If there is time, we will see how in week 5.

7.6 From the theorem to an argument for probabilism

The Dutch book theorem is a piece of mathematics. To get an argument for probabilism out of it we need some further premises, and the obvious way of supplying them is not very good.

The flat-footed version runs: if your credences are non-probabilistic, a cunning Dutchman might come along and take your money; if they are probabilistic, he cannot; so you had better have probabilistic credences.

Two objections to that:

- There might equally be a Frenchman going around *rewarding* people with non-probabilistic credences. We would not conclude from that that you ought to have them. At most we would conclude that having them would be practically useful.
- And that is the general shape of the problem: losing money is a practical misfortune, and what was supposed to be at issue is whether your beliefs are *epistemically* in order.

Here is a better version:

(1) An agent is rationally required to bet in line with the betting interpretation, given the credences they have.

(2) If an agent has non-probabilistic credences and bets in line with the betting interpretation, then they are vulnerable to exploitation by a Dutch book.

(3) If an agent is vulnerable to exploitation by a Dutch book, then their credences are irrational.

∴ If an agent has non-probabilistic credences, then their credences are irrational.

Assume the agent has non-probabilistic credences. By (1) she bets in line with the betting interpretation, so by (2) she is vulnerable to a Dutch book, and so by (3) her credences are irrational.

7.7 The Frenchman

BDRC objects that there might equally be a Frenchman going around *rewarding* people with non-probabilistic credences — so all the Dutch book shows is that probabilistic credences are practically useful.

I don't think that is a good refutation. The Frenchman has to turn up, and he rewards you for *having* the belief, never mind what it licenses you to do. Nobody has to turn up for the Dutch book: your own credences license every one of those bets, and together they lose you money for sure. Making choices which predictably leave you worse off, for no reason, is irrational.

7.8 And it isn't about money

It is not really about money. You are not rationally required to value money. But the argument goes through for any quantity you do care about, so long as you are required to bet in line with the betting interpretation for it.

7.9 Problems with the betting interpretation

The weak premise is the first one. You are obviously not required to maximise expected money, so you are not in general required to bet in line with the betting interpretation.

There is an answer. It is much more plausible that you are required to bet in line with the betting interpretation when the stakes are *small*. And no matter how small the stakes are made, you can still be made to lose for sure.

8 Readings for next week

Next week: **Newcomb's problem**.

- BDRC ch. 9.
- Adam Elga, "Newcomb University: A Play in One Act", *Analysis* 80.2 (2020), pp. 212–221.